VIT Bhopal University
School of Computing Science & Engineering

Project Title

Budget Friendly Travel Planner

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The **Budget Friendly Travel Planner** is a web-based application developed using Python and Flask that helps users plan affordable trips efficiently. The system allows users to enter their destination, travel dates, and budget, and generates a personalized travel plan including transportation options, accommodation suggestions, food recommendations, tourist attractions, and activities.

The main objective of this project is to simplify travel planning while maintaining cost efficiency. Many travellers face difficulties organizing trips within limited budgets. This application provides structured recommendations that assist users in making informed travel decisions.

The project demonstrates practical implementation of web development concepts, backend logic design, and basic AI-inspired recommendation systems. It also introduces students to full-stack development using modern technologies.

INTRODUCTION

Travel planning is often time-consuming and complicated, especially when budget constraints are involved. Students, families, and solo travelers frequently struggle to balance expenses while maximizing travel experiences.

The **Budget Friendly Travel Planner** aims to solve this problem by automating travel suggestions based on user inputs. The application integrates frontend user interaction with backend processing to produce optimized travel recommendations.

This project was developed as part of first-year learning to understand:

- Web application development
- Backend programming using Python
- User interface design
- Data handling using APIs

Being a student of **CSE (AI & ML)**, this project also introduces the concept of intelligent recommendation systems which can later be expanded using Artificial Intelligence.

OBJECTIVES OF THE PROJECT

The main objectives are:

- To develop a user-friendly travel planning system
- To provide budget-based travel recommendations
- To simplify trip organization
- To learn Flask web framework
- To integrate frontend and backend technologies
- To design a modular software structure
- To understand real-world problem solving using programming

PROBLEM STATEMENT

Planning a trip manually requires searching multiple websites for transportation, accommodation, food options, and tourist places. This process is inefficient and often exceeds the planned budget.

There is a need for a centralized system that:

- Accepts user travel requirements
- Generates structured travel plans
- Suggests affordable options
- Saves time and effort

The proposed system addresses these issues by providing an automated travel planning platform

METHODOLOGY

The system follows a client-server architecture:

Step 1: User Input

Users enter:

- Destination
- Budget
- Travel Dates

Step 2: Backend Processing

Flask backend processes user data and calls modules:

- destination.py
- accommodation.py
- transport.py
- food.py
- activities.py

Step 3: Plan Generation

The system generates:

- Transportation suggestions
- Accommodation options
- Food recommendations
- Activities
- Day-wise itinerary

Step 4: Result Display

The generated plan is returned in JSON format and displayed dynamically on the webpage.

SYSTEM ARCHITECTURE

Technologies Used

Frontend

- HTML
- CSS
- JavaScript

Backend

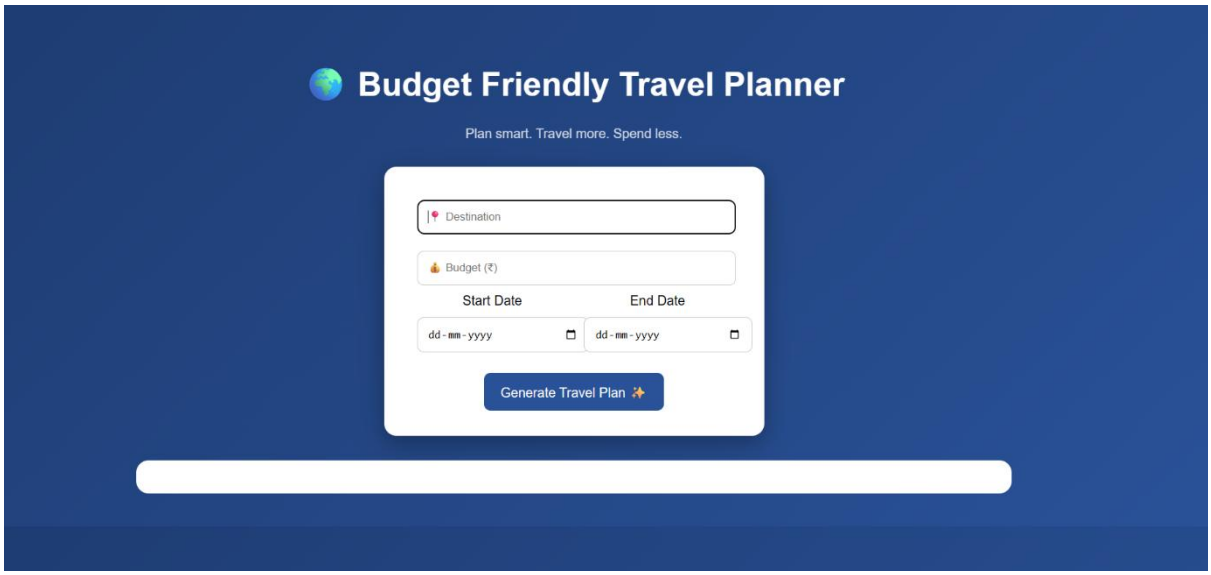
- Python
- Flask

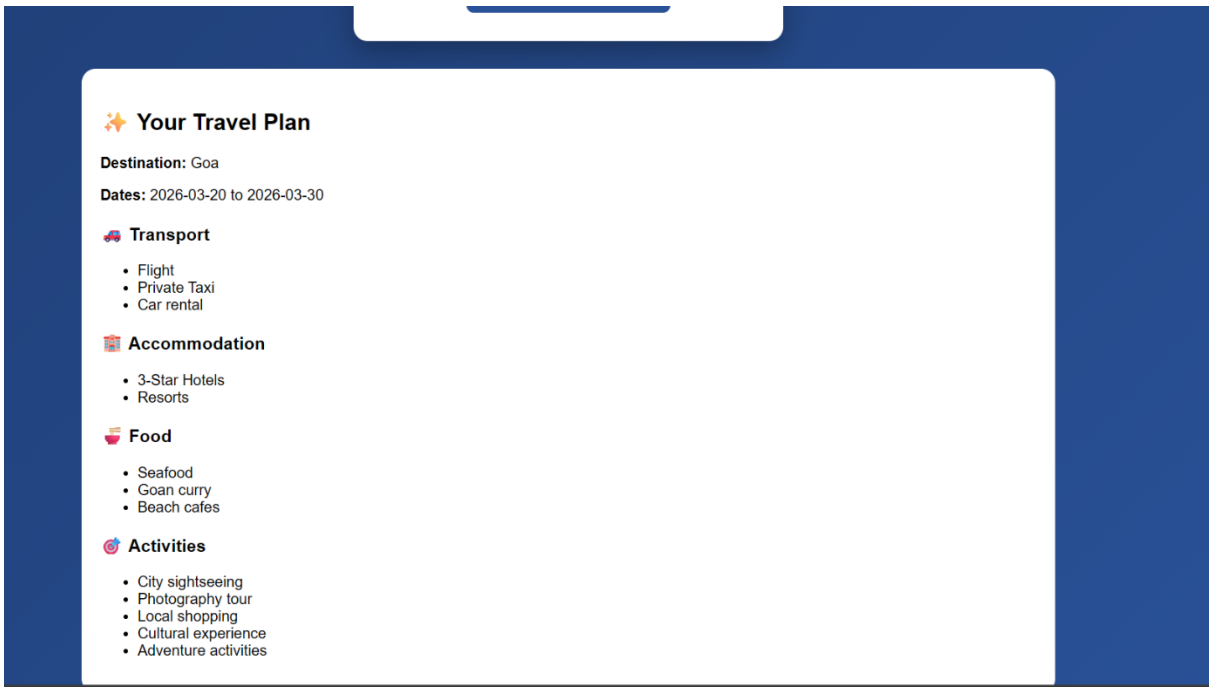
Architecture

User → Web Interface → Flask Server → Travel Modules → Generated Plan

FEATURES OF THE PROJECT

- Budget-based travel planning
- Dynamic itinerary generation
- Simple and interactive UI
- Modular backend design
- Real-time result generation
- Beginner-friendly architecture
- Expandable AI integration capability





APPLICATIONS & BENEFITS

This project can help:

- Students planning budget trips
- Solo travellers
- Families organizing vacations
- Beginners learning web development
- Future AI travel recommendation systems

Benefits include:

- Time saving
- Cost optimization
- Organized travel planning
- Easy accessibility

FUTURE ENHANCEMENTS

The project can be improved by adding:

- AI-based itinerary generation
- Machine learning recommendation engine
- Google Maps integration
- Hotel booking APIs
- User login system
- Cloud deployment
- Mobile application version

CONCLUSION

The Budget Friendly Travel Planner successfully demonstrates how programming concepts can solve real-world problems. The project helped in understanding web development, backend processing, and application design.

As a first-year **CSE AIML** student at VIT Bhopal, this project provided practical exposure to software development and laid the foundation for future AI-driven systems.

The application can be further enhanced using machine learning models to provide smarter and personalized travel recommendations.